

Drawing the Last Straw: Competition and Deception in a Duopoly

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Abstract. Consumers reward deception, such as hidden fees and inflated claims, up to a personal limit and abandon a firm that crosses it, yet firms rarely know where that limit lies. We study a duopoly in which consumers buy from the most deceptive firm they still tolerate. When firms know how tolerant consumers are, competition escalates deception: in the mixed-strategy equilibrium, average deception exceeds the monopoly level by about 41 per cent. When they do not, uncertainty cuts both ways. If a tolerant market is unlikely, firms deceive less on average, and earn less, than if they knew. If it is likely enough, firms gamble on it, and consumers periodically abandon a firm outright. These collapses are an equilibrium outcome, and they are most likely when firms are fairly, but not fully, confident in their customers' tolerance.

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JEL Classification: C72, D18, D43, D81, D91, L13

1. Introduction

Much of what firms do to sell a product leaves the product itself unchanged. A ticketing site advertises a low price and adds its fees at checkout; a shopping site runs a countdown timer that resets whenever the page is reloaded; a subscription takes one click to start and a string of screens to cancel. Practices like these are common: a crawl of roughly 11,000 shopping websites found manipulative interface designs, known as dark patterns, on about one in nine of them ([Mathur et al., 2019](#)). Regulators have begun to respond. In 2023, India's Central Consumer Protection Authority prohibited thirteen specified types of dark pattern ([Central Consumer Protection Authority, 2023](#)), and in 2025 Amazon agreed to pay \$2.5 billion to settle charges by the U.S. Federal Trade Commission that it had enrolled consumers in its Prime subscription without their consent and made the subscription hard to cancel ([Federal Trade Commission, 2025](#)).

Three features of such deception motivate this paper. The first is that, up to a point, it works. Deception makes an offer look better, and consumers respond to the offer as it looks: hiding part of a price raises demand ([Chetty et al., 2009](#); [Blake et al., 2021](#)), and in a large online experiment, mild dark patterns more than doubled the share of subjects who signed up for a dubious data-protection plan ([Luguri and Strahilevitz, 2021](#)). The second is that, beyond that point, it backfires. In the same experiment, aggressive dark patterns raised sign-ups further but also provoked a strong backlash, which mild ones did not. More broadly, consumers judge

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some profit-seeking conduct to be unfair (Kahneman et al., 1986), and customers who feel short-changed by a firm go on to buy less from it (Anderson and Simester, 2010). Consumers, in short, tolerate deception only up to a limit, and the deception that crosses it is the proverbial last straw. The third is that this limit is hard to predict. It differs across consumers: in the experiment above, less educated subjects were markedly more susceptible to mild dark patterns than their better-educated counterparts. And how tolerant consumers are as a whole can differ from one market to the next, so a firm introducing a new fee or a new interface design cannot know in advance how much its own customers will bear. Every such decision is, in part, a bet on how tolerant they turn out to be.

This paper asks how competition shapes deception when consumers behave this way, and what changes when firms do not know how much their consumers will tolerate. One common presumption is that rivalry disciplines deception, since a firm that deceives less can win customers from one that deceives more. Akerlof and Shiller (2015) argue the opposite: competitive pressure ensures that every profitable opportunity to manipulate consumers is taken. We show that in markets like these, competition does more than sustain deception. It escalates it: duopolists deceive more than a monopolist would, on average by a wide margin. Not knowing consumers' limits then cuts both ways. When firms think it unlikely that consumers are very tolerant, uncertainty restrains deception. When they think it likely enough, it turns deception into a gamble, and a firm that loses the gamble loses its entire market.

We study a duopoly in which two symmetric firms simultaneously choose a level of deception, meaning any practice that raises the perceived attractiveness of a product without changing the product itself. Each consumer has a private tolerance threshold, and thresholds are uniformly distributed up to a ceiling θ . Below her threshold, a consumer does not register deception as deception; she sees a more attractive offer. Above it, she rejects the firm outright. Each consumer therefore buys from the most deceptive firm she still tolerates. Deception then moves a firm's profit through three channels. It raises the revenue extracted from each retained customer. It drives away the marginal customer whose threshold it crosses. And, under competition only, it lets a firm overtake its rival and serve every consumer tolerant enough to accept the higher level. A monopolist weighs the first two channels against each other; a duopolist must weigh all three.

We first solve the game when firms know θ . The overtaking channel rules out equilibrium in pure strategies for any continuous, strictly increasing revenue function (Proposition 1). If both firms choose the same level, either can take the whole top of the market, rather than half of it, by deceiving slightly more. If they choose different levels below the ceiling, the less deceptive firm, which serves only the consumers its rival drives away, gains by leapfrogging. Firms therefore randomize, and otherwise identical firms in the same market end up deceiving to different degrees. The logic mirrors that of price dispersion in Varian (1980), except that the most price-sensitive consumers there flock to the firm that undercuts, while the most tolerant consumers here flock to the firm that overtakes. Just as randomized prices can be read as occasional sales, randomized deception can be read as firms alternating between aggressive and restrained tactics, or as dispersion in tactics across the sellers in a market.

We characterize a symmetric equilibrium in which each firm draws its deception level from a continuous distribution on an interval. Given the two endpoints of the interval, the density has a closed form, and the endpoints themselves have a clear economic logic. A firm at the bottom of the interval is always the less deceptive of the two, so it chooses its level optimally against its

rival's average deception; with iso-elastic revenue $r(d) = d^\gamma$, this implies that mean deception is exactly $(\gamma + 1)/\gamma$ times the lowest level played. A firm at the top of the interval is always the more deceptive of the two and earns what a monopolist would at that level, which places the top of the interval above the monopoly level but strictly below θ . The equilibrium also scales with the ceiling (Proposition 3), so every comparison with monopoly is independent of how tolerant consumers are.

Our first main result is that competition pushes deception above the monopoly level. With linear revenue, average deception in equilibrium exceeds the monopoly level by about 41 per cent and the highest level played exceeds it by about 71 per cent, while each firm earns just under half of the monopoly profit (Corollary 1). The escalation is largest precisely where revenue is least responsive to deception. As the revenue elasticity of deception γ falls toward zero, a monopolist stops deceiving, because deception barely raises what it extracts from each customer. Duopolists keep deceiving, because overtaking a rival pays through market share rather than through revenue per customer, so the ratio of duopoly to monopoly deception grows without bound (Table 1 and Figure 1).

The second half of the paper, and its main focus, turns to incomplete information (Section 5). The market is either fragile, with ceiling θ_F , or robust, with a higher ceiling θ_R , and firms choose deception without observing which. Both firms know the probability μ that the market is robust, and neither has private information. A firm that deceives beyond θ_F is therefore gambling on the robust market: if the market turns out to be fragile, every consumer rejects it and it sells nothing. The analysis rests on a simple observation. Each firm's demand is linear in the distribution of consumer tolerance, so expected payoffs under uncertainty are exactly the payoffs in a known market whose tolerance distribution is the probability-weighted average of the two. The tools of the complete information game therefore carry over to the uncertain one.

When a robust market is unlikely enough, firms never deceive past θ_F , and the equilibrium is exactly the complete information one for an effective ceiling: the harmonic mean of θ_F and θ_R , weighted by the probabilities of the two states (Proposition 4). The harmonic mean appears because what matters to a firm is not the ceiling itself but how quickly it loses customers as it deceives more, and it is these rates, rather than the ceilings, that uncertainty averages. Since a harmonic mean lies below the corresponding arithmetic mean, uncertainty restrains deception. The whole distribution of deception shifts down relative to a market known to have the average ceiling, and firms deceive less on average, and earn less, than they would if they could learn how tolerant consumers are before choosing. The restraint is larger the more uncertain firms are and the more the two markets differ.

When a robust market is likely enough, this caution breaks down and firms gamble: each puts positive probability on deception beyond θ_F . Whenever the market then turns out to be fragile, an overreaching firm loses every customer, an event we call a bust. Busts are an equilibrium outcome rather than a mistake: firms know the odds and accept the risk because the gamble pays on average. The probability of a bust is hump-shaped in firms' confidence (Figure 2b). It is zero when a robust market is unlikely, since firms are then cautious, and it vanishes as a robust market becomes certain, since a fragile market then almost never occurs. In between it can be large: in one of our examples, the probability that at least one firm busts reaches 58 per cent. How firms gamble depends on how much more tolerant the robust market is (Figure 2a). If it is only moderately more tolerant, firms' deception ranges continuously across θ_F , and they

sometimes overreach by only a little. If it is much more tolerant, each firm’s strategy splits in two: it either stays safely below θ_F or deceives far beyond it, and levels near θ_F are never played. The reason is that crossing θ_F makes further deception cheaper, since the fragile market’s customers are already lost, so a firm that crosses θ_F has reason to go well beyond it. And when the robust market is both much more tolerant and likely, firms go all in: they ignore the fragile market altogether, and whenever it turns out to be fragile, the whole market collapses.

These results carry several implications for policy. First, where deception works by inflating perceived value, competition is no substitute for consumer protection: adding a rival raises the average level of deception rather than lowering it. Nor will consumers’ own choices correct the escalation. By assumption, deception raises the utility on which consumers choose while lowering the utility they eventually experience (Kahneman et al., 1997), so purchase decisions respond to the utility that deception inflates rather than to the utility it erodes. Second, because equilibrium deception is proportional to the tolerance ceiling, measures that make consumers more vigilant, and so lower θ , reduce deception under monopoly and duopoly alike, but they leave the escalation caused by competition untouched. Third, where firms are cautious, better information about consumers raises deception: firms deceive more on average, and earn more, when they know how much their customers will tolerate than when they do not. Tools that let firms learn their customers’ limits, such as experimenting with interface designs on live traffic, therefore push deception up. Fourth, episodes in which consumers abandon a firm need not mean that the firm misjudged them. In our model, they are the downside of bets that firms rationally make, and they are most frequent not when firms are reckless but when they are fairly, though not fully, confident that consumers will tolerate more.

1.1. Related literature

This paper contributes to behavioral industrial organization, and in particular to the study of deception in competitive markets; Spiegler (2011) and Heidhues and Kőszegi (2018) survey the field. A recurring theme is that competition need not eliminate deception. In Gabaix and Laibson (2006), firms keep add-on prices shrouded because educating consumers does not pay: a consumer who learns to avoid the add-on has no reason to reward the firm that taught her. Ellison (2005) shows how add-on pricing can soften price competition, and Heidhues et al. (2017) show that profitable deception can keep inferior products in business even in otherwise competitive markets. Models of obfuscation go further and find that competition can increase the complexity firms impose on consumers. Carlin (2009) shows that firms may respond to additional competitors by making their prices more complex instead of more transparent, and Spiegler (2016) finds that this pattern runs through much of the literature. We identify a different route to a similar conclusion. In obfuscation models, complexity pays because it keeps consumers from comparing offers and so softens price competition. Here there are no prices to compete on: deception is itself the instrument of competition, because consumers are drawn to it up to their tolerance. The benchmark is correspondingly sharp. A monopolist facing the same consumers and the same revenue from deception deceives less, on average, than either duopolist.

The closest setting to ours is that of Glaeser and Ujhelyi (2010), who study firms that spend to misinform consumers about the health costs of a product. When misinformation shifts beliefs about a whole product category, it is a public good among firms, and they show that it falls as the number of firms rises. In an extension, they note that when misinformation is firm-specific,

so that every consumer buys from whichever firm has misled consumers most, simultaneous choices admit no equilibrium in pure strategies, and they turn to a sequential game instead. We study a simultaneous-move game of this firm-specific kind, with one important difference: because every consumer has a limit, the less deceptive firm is not shut out of the market but keeps the consumers its rival drives away. Set side by side, the two models suggest that whether competition dilutes or sharpens the incentive to deceive depends on whether deception expands the whole market or takes customers from rivals.

The tolerance thresholds formalize the idea that consumers punish conduct they perceive as exploitative. [Kahneman et al. \(1986\)](#) document that the public judges some profit-seeking actions to be unfair; [Rotemberg \(2005\)](#) models consumers who react against a price increase only once they become convinced it is unfair; and [Anderson and Simester \(2010\)](#) show in a field experiment that customers who see a retailer cut the price of an item they recently bought go on to buy less from it. Our consumers behave similarly, with one addition: below their threshold, they do not merely tolerate deception but are attracted by it.

Our treatment of incomplete information is deliberately simple. Following [Harsanyi \(1967\)](#), we model the firms' ignorance as a move by nature, but because neither firm has private information, each firm has a single type and the game reduces to one in expected payoffs. Much of the work on incomplete information in oligopoly instead gives firms private signals about demand or costs and asks whether they gain by sharing them ([Vives, 1984](#); [Raith, 1996](#)). Common uncertainty lets us isolate a different question: not what firms learn from each other, but how not knowing consumers' limits changes the incentive to deceive. The answer turns on what uncertainty averages. Because it averages the rates at which deception loses customers rather than the ceilings themselves, it first makes firms more cautious than they would be in the average market, and then, once a tolerant market is likely enough, leads them to gamble on it.

Methodologically, the equilibrium belongs to the family of mixed-strategy equilibria familiar from models of price dispersion ([Varian, 1980](#); [Burdett and Judd, 1983](#)). Games of this kind have discontinuous payoffs, the setting studied by [Dasgupta and Maskin \(1986\)](#); here we construct the equilibrium directly. The difference lies in the direction of competition. In models of price dispersion, a firm wins customers by undercutting, and prices are dispersed below the monopoly price. Here a firm wins customers by overtaking, and deception is dispersed around the monopoly level, lying above it on average. Under incomplete information, the support of the equilibrium can also split into two separate intervals, a feature we trace to a kink in expected demand at the fragile market's ceiling.

The rest of the paper is organized as follows. Section 2 sets out the model. Section 3 solves the monopoly benchmark and shows that the duopoly has no equilibrium in pure strategies. Section 4 characterizes the mixed-strategy equilibrium under complete information and its comparative statics. Section 5 turns to incomplete information: it shows that an uncertain market is equivalent to a known one with an averaged tolerance distribution, characterizes when firms are cautious and what uncertainty then does to deception, and maps when and how firms gamble.

2. The Model

We consider a static duopoly game of incomplete information. Two symmetric firms, indexed by $i \in \{1, 2\}$, compete by simultaneously choosing a level of strategic deception $d_i \in [0, \bar{d}]$, where $\bar{d}$

represents the structural upper bound of feasible deceptive practices.

2.1. Consumer Behavior and Demand Microfoundations

The market consists of a continuum of heterogeneous consumers normalized to a total mass of one. 'Deception' in this market is defined as an action taken by a firm that increases the perceived attractiveness of a product to consumers without intrinsically changing the product itself. Each consumer is characterized by a private tolerance threshold τ , which represents the maximum level of deception they will tolerate before completely rejecting a firm's product. We assume that consumer tolerance is uniformly distributed across the population up to an aggregate market limit θ , such that $\tau \sim U[0, \theta]$.

A consumer facing a firm playing deception level d_i evaluates their options according to their decision utility:

Assumption 1 (Consumer Decision Utility). The utility a consumer with private tolerance τ derives from purchasing from firm i is given by

$$\tilde{U}(d_i, \tau) = \begin{cases} v(d_i) & \text{if } d_i \leq \tau \\ 0 & \text{if } d_i > \tau \end{cases} \quad (1)$$

with $v'(d_i) > 0$. Since we only model consumer choice, experienced utility is not modeled directly. However, we assume that it is a decreasing function in deception, i.e. deception inflates the perceived value of the product when making the choice to buy, but reduces actual experienced utility once the consumer eventually realizes they've been deceived.

A higher level of d_i artificially inflates the product's perceived attractiveness, driving up the consumer's perceived utility. This psychological illusion persists continuously until the firm overreaches and crosses the consumer's private tolerance threshold τ . At this point, the consumer detects the deception and ceases to buy from the firm.

Let $d_H = \max\{d_1, d_2\}$ and $d_L = \min\{d_1, d_2\}$ denote the high and low deception firms. Given that consumers maximize utility subject to their tolerance constraint, the aggregate demand schedules q_H and q_L are determined by the realization of θ .

$$q_H(d_H; \theta) = \begin{cases} \frac{\theta - d_H}{\theta}, & d_H \leq \theta, \\ 0, & d_H > \theta, \end{cases} \quad q_L(d_L, d_H; \theta) = \begin{cases} \frac{d_H - d_L}{\theta}, & d_H \leq \theta, \\ \frac{\theta - d_L}{\theta}, & d_L \leq \theta < d_H, \\ 0, & d_L > \theta. \end{cases} \quad (2)$$

2.2. Firm Payoffs and the Optimization Problem

Firms are risk-neutral and maximize expected profits. Deception operates as a double-edged sword: it reduces consumer demand, but increases the revenue extracted per retained customer. We formalize this revenue a firm can extract *because* deception inflates perceived value as $r(d_i) = d_i^\gamma$, where $\gamma > 0$. We will make two useful assumptions on r :

1. $r(0) = 0$, and r is positive, differentiable and strictly increasing on $(0, \bar{d}]$,

2. $\frac{r(d)^2}{r'(d)}$ is strictly increasing on $(0, \bar{d}]$.

Firms face no production costs. The profit function for firm i is given by:

$$\pi_i(d_i, d_j; \theta) = r(d_i) \cdot q_i(d_i, d_j; \theta) \quad (3)$$

3. Solving the model with complete information

3.1. Monopoly benchmark

A single firm in this environment would maximize $\pi = d^\gamma \frac{\theta-d}{\theta}$. Maximizing this, we get the result

$$d^{mon} = \frac{\gamma}{\gamma+1} \theta, \quad \pi^{mon} = \frac{\gamma^\gamma}{(\gamma+1)^{\gamma+1}} \theta^\gamma$$

3.2. Pure Strategy Nash Equilibrium

Proposition 1. *For any strictly increasing continuous $r(d)$, the game has no pure strategy Nash equilibrium.*

Proof. If $d_1 = d_2 = d$ and $d < \theta$, both firms share the consumers tolerant to d equally, each earning $\pi_i = r(d) \frac{\theta-d}{2\theta}$. Consider a deviation to $d + \varepsilon$. Profit for the deviator now becomes $\pi_i = r(d + \varepsilon) \frac{\theta-d-\varepsilon}{\theta}$ as they no longer have to share the market. As $\varepsilon \rightarrow 0$, $\pi_i = r(d) \frac{\theta-d}{\theta}$, which is twice as high. Therefore, a small positive deviation is strictly profitable.

If $d_L < d_H$ and $d_H > \theta$, the high firm earns zero and can deviate to below θ to earn positive profits. If $d_H < \theta$, we know that at equilibrium, $\pi_H(d_H) > \pi_H(d) \forall d$. In particular, $\pi_H(d_H) > \pi_H(d_L)$ for the high firm. Now for the low firm, it earns $\pi_L = r(d_L) \frac{d_H-d_L}{\theta}$. But notice this is strictly less than the numerator of $\pi_H(d_L) = r(d_L) \frac{\theta-d_L}{\theta}$ as $d_H < \theta$. Therefore, we get $\pi_H(d_H) > \pi_H(d_L) > \pi_L(d_L)$. If the low firm plays $d_H + \varepsilon$, it becomes the new high firm and earns a higher profit than being the low firm, giving it an incentive to deviate. $\square$

4. Mixed Strategy Nash Equilibrium

4.1. Indifference condition

Suppose the firm earns the same π^* across the entire support $[\underline{s}, \bar{s}]$. If $F(d)$ denotes the CDF from which the firms draw actions, then their expected profit becomes

$$\mathbb{E}(\pi) = r(d) \left[F(d) \frac{\theta-d}{\theta} + (1-F(d)) \mathbb{E} \left[\frac{x-d}{\theta} | x > d \right] \right] = \pi^* \quad \forall d \in [\underline{s}, \bar{s}] \quad (4)$$

To solve for MSNE, we need to set $\mathbb{E}'(\pi) = 0$. We'll do so by differentiating elements of $\mathbb{E}(\pi)$ one by one. If $f(d)$ is the density of $F(d)$, we can rewrite the conditional expectation. Let's call it the expected residual demand, represented by $R(d)$.

$$(1-F(d)) \mathbb{E} \left[\frac{x-d}{\theta} | x > d \right] = \int_d^{\bar{s}} \frac{x-d}{\theta} f(x) dx \equiv R(d)$$

4.2. Solving for the density

Differentiating it, we get

$$\begin{aligned}
\frac{d}{dd} \int_d^{\bar{s}} \frac{x-d}{\theta} f(x) dx &= \frac{1}{\theta} \left[-(d-d)f(d) + \int_d^{\bar{s}} \frac{\partial}{\partial d} [(x-d)f(x)] dx \right] \\
&= \frac{1}{\theta} \left[\int_d^{\bar{s}} -(f(x)) dx \right] \\
&= \frac{1}{\theta} [-(F(\bar{s}) - F(d))] \\
\implies R'(d) &= \frac{-(1 - F(d))}{\theta}
\end{aligned}$$

Next, differentiating $F(d)\frac{\theta-d}{\theta} + R(d)$ gives us

$$\begin{aligned}
\frac{d}{dd} \left[F(d)\frac{\theta-d}{\theta} + R(d) \right] &= f(d)\frac{\theta-d}{\theta} - \frac{F(d)}{\theta} + R'(d) \\
&= f(d)\frac{\theta-d}{\theta} - \frac{F(d)}{\theta} - \frac{1-F(d)}{\theta} \\
&= \frac{f(d)(\theta-d)}{\theta} - \frac{1}{\theta}
\end{aligned} \tag{5}$$

Finally, differentiating $\mathbb{E}(\pi)$ yields

$$\mathbb{E}'(\pi) = r'(d) \left[F(d)\frac{\theta-d}{\theta} + R(d) \right] + r(d) \left[\frac{f(d)(\theta-d) - 1}{\theta} \right] = 0$$

Since $\left[F(d)\frac{\theta-d}{\theta} + R(d) \right] = \frac{\pi^*}{r(d)}$, we can clean up this expression to get

$$\begin{aligned}
\mathbb{E}'(\pi) &= \frac{r'(d)\pi^*}{r(d)} + \frac{r(d)}{\theta} [f(d)(\theta-d) - 1] = 0 \\
\implies \frac{r'(d)\pi^*\theta}{r(d)^2} + f(d)(\theta-d) - 1 &= 0
\end{aligned}$$

Therefore, we get the density

$$f(d) = \frac{1}{\theta-d} \left[1 - \frac{r'(d)\pi^*\theta}{r(d)^2} \right] \tag{6}$$

4.3. Identifying the optimal profit

Our next goal will be to pin down π^* . To do so, we can evaluate it at $F(\underline{s})$ and $F(\bar{s})$, values of F we know. At $d = \underline{s}$, $F(\underline{s}) = 0$, giving us

$$\begin{aligned}\pi^* &= r(\underline{s}) \int_{\underline{s}}^{\bar{s}} \frac{x - \underline{s}}{\theta} f(x) dx \\ &= \frac{r(\underline{s})}{\theta} \left[\int_{\underline{s}}^{\bar{s}} x f(x) dx - \underline{s} \int_{\underline{s}}^{\bar{s}} f(x) dx \right] \\ &= \frac{r(\underline{s}) [\mathbb{E}[d] - \underline{s}]}{\theta}\end{aligned}$$

For any $d < \underline{s}$, we know that $F(d) = 0$, so we get a payoff of the same form $\pi(d) = \frac{r(d) [\mathbb{E}[d] - d]}{\theta}$. Since this d is below the support, it is not in the optimal interval for the MSNE, and thus it must earn a strictly lower profit than $\pi(\underline{s}) = \pi^*$. In other words, $\underline{s}$ maximizes profits, and thus also maximizes $\frac{r(d) [\mathbb{E}[d] - d]}{\theta}$ for $d \leq \underline{s}$. Differentiating, the FOC is $r'(\underline{s}) [\mathbb{E}(d) - \underline{s}] - r(\underline{s}) \geq 0$ at the right side boundary, and thus we get

$$\mathbb{E}[d] - \underline{s} \geq \frac{r(\underline{s})}{r'(\underline{s})}$$

Substituting into the expression for π^* , we get

$$\pi^* \geq \frac{r^2(\underline{s})}{r'(\underline{s})\theta}$$

But from the density (Equation 6), $f(\underline{s}) \geq 0 \implies \frac{r^2(\underline{s})}{r'(\underline{s})\theta} \geq \pi^*$. Combining the two inequalities, we get

$$\pi^* = \frac{r^2(\underline{s})}{r'(\underline{s})\theta}$$

At $d = F(\bar{s}) = 1$, $\pi^* = r(\bar{s}) \frac{\theta - \bar{s}}{\theta}$, so

$$\pi^* = \frac{r(\bar{s})(\theta - \bar{s})}{\theta}$$

Since $\pi^* > 0$, this establishes that the upper end of the support $\bar{s} < \theta$, so firms never deceive past the market boundary in a complete information MSNE.

Proposition 2 (Expected value of deception). *Under complete information, $\mathbb{E}[d] = \frac{\gamma+1}{\gamma}\underline{s}$.*

Proof. We know $\pi^* = \frac{r(\underline{s})[\mathbb{E}[d] - \underline{s}]}{\theta} = \frac{r^2(\underline{s})}{r'(\underline{s})\theta}$. This gives us $\mathbb{E}[d] = \underline{s} + \frac{r(\underline{s})}{r'(\underline{s})} = \frac{\gamma+1}{\gamma}\underline{s}$. $\square$

4.4. With constant elasticity revenue

To move further, let's use our assumed functional form $r(d) = d^\gamma$. This gives us

$$\pi^* = \frac{\underline{s}^{\gamma+1}}{\gamma\theta} = \frac{\bar{s}^\gamma(\theta - \bar{s})}{\theta}$$

and our density is

$$f(d) = \frac{1}{\theta - d} \left[1 - \left(\frac{\underline{s}}{d} \right)^{\gamma+1} \right]$$

Integrating over the whole support gives us

$$\int_{\underline{s}}^{\bar{s}} \frac{1}{\theta - d} \left[1 - \left(\frac{\underline{s}}{d} \right)^{\gamma+1} \right] dd = 1$$

For simplicity, let $\gamma = 1$. Then, $f(d) = \frac{1}{\theta - d} - \frac{\underline{s}^2}{d^2(\theta - d)}$. Integrating this gives us

$$F(d) = \left(\left(\frac{\underline{s}}{\theta} \right)^2 - 1 \right) \ln(\theta - d) - \left(\frac{\underline{s}}{\theta} \right)^2 \ln(d) + \left(\frac{\underline{s}}{\theta} \right)^2 \frac{\theta}{d} + C$$

Since we know $F(\bar{s}) - F(\underline{s}) = 1$, subtracting gives us

$$(\rho^2 - 1) \ln \left(\frac{\theta - \bar{s}}{\theta - \underline{s}} \right) - \rho^2 \ln \left(\frac{\bar{s}}{\underline{s}} \right) + \rho^2 \theta \left(\frac{1}{\bar{s}} - \frac{1}{\underline{s}} \right) = 1$$

where $\rho \equiv \frac{\underline{s}}{\theta}$. From π^* , we get $\underline{s}^{\gamma+1} = \gamma \bar{s}^\gamma (\theta - \bar{s})$. Therefore, the above expression is an equation in one variable which can be solved numerically. Doing so for $\theta = 1$, we get $\bar{s} = 0.8557$ and thus $\underline{s} = 0.3514$.

Proposition 3 (Scale invariance). *Fix $\gamma > 0$ and let $\theta > 0$ be arbitrary. If F_1 is an equilibrium distribution for a market with tolerance ceiling 1, then the distribution F_θ defined by $F_\theta(d) = F_1(d/\theta)$ is an equilibrium distribution for a market with ceiling θ , and conversely. Consequently*

$$\underline{s}(\theta) = \theta \underline{s}(1), \quad \bar{s}(\theta) = \theta \bar{s}(1), \quad \mathbb{E}_\theta[d] = \theta \mathbb{E}_1[d], \quad \pi^*(\theta) = \theta^\gamma \pi^*(1).$$

Proof. Write actions as $d_i = \theta u_i$ with $u_i \in [0, 1]$. Since the demand schedules depend on deception only through its ratio to the ceiling,

$$q_H = \frac{\theta - d_H}{\theta} = 1 - u_H, \quad q_L = \frac{d_H - d_L}{\theta} = u_H - u_L,$$

so demand in the market with ceiling θ evaluated at $(\theta u_1, \theta u_2)$ equals demand in the market with ceiling 1 evaluated at (u_1, u_2) . Revenue is homogeneous of degree γ , since $r(\theta u) = (\theta u)^\gamma = \theta^\gamma r(u)$. Therefore

$$\pi_i(\theta u_i, \theta u_j; \theta) = r(\theta u_i) q_i(u_i, u_j) = \theta^\gamma \pi_i(u_i, u_j; 1).$$

The map $d \mapsto d/\theta$ is thus a bijection between the action spaces under which every payoff in the market with ceiling θ equals θ^γ times the corresponding payoff in the market with ceiling 1. $\square$

Corollary 1 (Competitive escalation). *Let $\gamma = 1$. In the mixed strategy equilibrium, for every tolerance ceiling θ ,*

$$\underline{s} = 0.3514\theta, \quad \mathbb{E}[d] = 0.7029\theta, \quad \bar{s} = 0.8557\theta,$$

and, relative to the monopoly benchmark $d^{\text{mon}} = \gamma\theta/(\gamma + 1)$,

$$\frac{\mathbb{E}[d]}{d^{\text{mon}}} = 1.406, \quad \frac{\bar{s}}{d^{\text{mon}}} = 1.711, \quad \frac{\pi^*}{\Pi^{\text{mon}}} = 0.494.$$

These ratios are independent of θ . Average deception under duopoly therefore exceeds the monopoly level by roughly 41 per cent, and the highest level played exceeds it by roughly 71 per cent, while equilibrium profit falls to under half the monopoly benchmark.

Proof. The solution of the normalisation condition at $\theta = 1$ gives $\bar{s} = 0.8557$, and $\underline{s} = \sqrt{\bar{s}(1 - \bar{s})} = 0.3514$; the mean follows from $\mathbb{E}[d] = (\gamma + 1)\underline{s}/\gamma$. Rescaling by Proposition 3 gives the stated levels for arbitrary θ . $\square$

As we sweep across various values of γ , the revenue elasticity of deception, we get the following results:

Table 1: Equilibrium deception across revenue elasticities

γ	Equilibrium distribution				Relative to monopoly		
	$\underline{s}$	$\mathbb{E}[d]$	$\bar{s}$	$\bar{s} - \underline{s}$	$\frac{\mathbb{E}[d]}{d^{\text{mon}}}$	$\frac{\bar{s}}{d^{\text{mon}}}$	$\frac{\pi^*}{\pi^{\text{mon}}}$
0.25	0.104	0.522	0.744	0.640	2.610	3.722	0.444
0.50	0.201	0.604	0.798	0.596	1.812	2.393	0.470
0.75	0.283	0.661	0.832	0.549	1.542	1.941	0.484
1.00	0.351	0.703	0.856	0.504	1.406	1.711	0.494
1.50	0.457	0.762	0.887	0.431	1.269	1.479	0.506
2.00	0.534	0.801	0.908	0.374	1.201	1.361	0.513
3.00	0.638	0.850	0.932	0.295	1.133	1.243	0.522
4.00	0.704	0.880	0.946	0.242	1.100	1.183	0.527
5.00	0.750	0.900	0.955	0.206	1.079	1.147	0.530

Figure 1a plots equilibrium deception relative to the monopoly benchmark, and Figure 1b the support of the equilibrium distribution, both as fractions of θ .

The ratio wrt monopoly falls in γ and diverges as $\gamma \rightarrow 0$. From Equation 5, the derivative of expected demand is $D'(d) = \frac{f(d)(\theta - d)}{\theta} - \frac{1}{\theta}$. It tells us that raising d forfeits the mass $1/\theta$ of consumers whose tolerance is just crossed, but overtakes any rival and becoming the high firm with probability $f(d)$. A monopolist faces the first effect and not the second, only deceiving for the added revenue from γ . As $\gamma \rightarrow 0$ the monopolist therefore stops deceiving, $d^{\text{mon}} \rightarrow 0$, while duopolists retain the incentive to leapfrog. Even though deception is increasing in γ , as the duopolists still retain some incentive to deceive while monopolists do not, the ratio $\frac{\mathbb{E}[d]}{d^{\text{mon}}}$ blows up as $\gamma \rightarrow 0$.

On the other hand, the support narrows as γ rises. This is because across the support, we must earn the same π^* . In our model, this works because the gain from increased deception perfectly cancels out the loss from lost market share on $[\underline{s}, \bar{s}]$. But as γ rises, revenue extracted

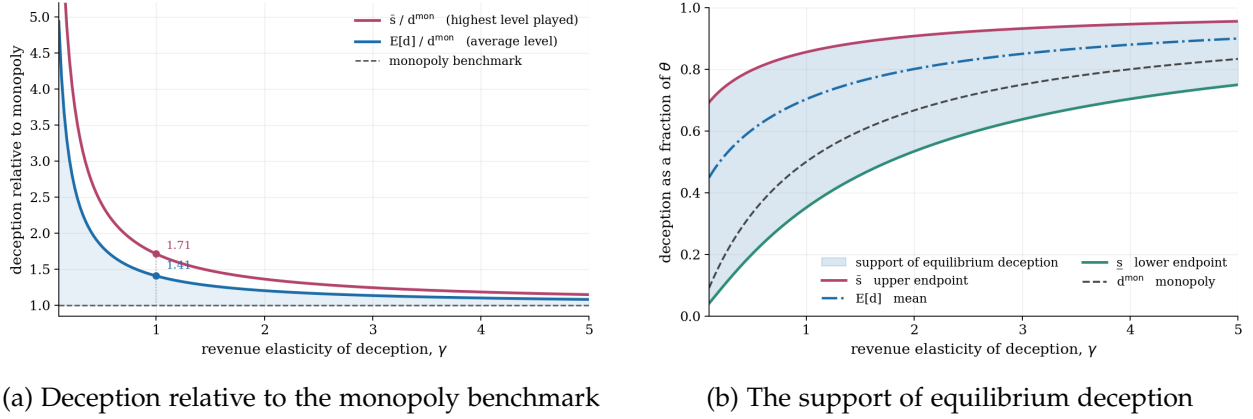


Figure 1: Comparative statics in the revenue elasticity γ .

increases exponentially. Thus, as γ rises, the revenue gains are so significant there may not be enough market share to compensate the lower end of the support, making it narrower.

5. Incomplete Information on Consumer Tolerance

So far, both firms knew exactly how much deception consumers would put up with. In practice, a firm rarely knows how far it can push consumers before they walk away. We model this here by making firms pick deception levels when they don't know the tolerance threshold θ .

5.1. Setup

There are two states of the market. Consumer tolerance is either *fragile* or *robust*, represented by θ_F and θ_R respectively, with $0 < \theta_F < \theta_R$. Let $\kappa \equiv \frac{\theta_R}{\theta_F} > 1$ measure how much more tolerant the robust market is. Nature draws the state, and the market is robust with probability $\mu \in (0, 1)$, and both firms know μ . This could be thought of as firms learning from past experience and cases how robust such markets tend to be, but they don't know the robustness of the current market they're playing in. Firms pick d_1 and d_2 simultaneously without observing the realization of θ . We assume that $\bar{d} \geq \theta_R$, so firms can deceive past the robust threshold if they so please. Firm i maximizes

$$\mathbb{E}_\theta[\pi_i(d_i, d_j; \theta)] = (1 - \mu)\pi_i(d_i, d_j; \theta_F) + \mu\pi_i(d_i, d_j; \theta_R)$$

5.2. Expected demand

Our tolerance distribution has the CDF $G(x)_\theta = \min\{\frac{x}{\theta}, 1\}$. Since the high firm serves consumers with $\tau \geq d_H$ and the low firm serves $d_L \leq \tau < d_H$, demand for each firm is

$$q_L = G_\theta(d_H) - G_\theta(d_L) \quad q_H = 1 - G_\theta(d_H)$$

Let $G_F \equiv G_{\theta_F}$ and $G_R \equiv G_{\theta_R}$ denote the CDF in the fragile and robust states respectively. Then,

expected demand can be written as

$$\begin{aligned}\mathbb{E}_\theta[q_H] &= (1 - \mu)[1 - G_F(d_H)] + \mu[1 - G_R(d_H)] = 1 - G(d_H) \\ \mathbb{E}_\theta[q_L] &= (1 - \mu)[G_F(d_H) - G_F(d_L)] + \mu[G_R(d_H) - G_R(d_L)] = G(d_H) - G(d_L)\end{aligned}$$

where G represents the belief weighted CDF

$$G(x) \equiv (1 - \mu)G_F(x) + \mu G_R(x) \quad (7)$$

Lemma 1 (Equivalence). *At every deception profile, a firm's expected profit in the game with uncertain tolerance ceiling equals its profit in the complete information game where the consumer tolerance has the CDF G as (7). Thus, the games have the same Nash equilibria.*

Proof. Since $r(d_i) = d_i^\gamma$ regardless of whether θ is robust or fragile, it acts as a constant when calculating expectations, so $\mathbb{E}_\theta[\pi_i] = r(d_i)\mathbb{E}_\theta[q_i]$. The above expressions show us that $\mathbb{E}_\theta[q_i]$ under incomplete information equals q_i in a complete information game when the consumer has the CDF $G(x)$. Thus, $\mathbb{E}_\theta[\pi_i] = \pi_i$ when $d_1 \neq d_2$. In a tie, each firm sells $\frac{1 - G_\theta(d)}{2}$, which in expectation is $\frac{1 - G(d)}{2}$, the same as a complete information game with CDF $G(x)$, giving $\mathbb{E}_\theta[\pi_i] = \pi_i$ again. Therefore, payoffs are always equal across the two games, thus we get the same equilibria. $\square$

Computing G piecewise, we get

$$G(x) = \begin{cases} x/\tilde{\theta}, & 0 \leq x \leq \theta_F \\ (1 - \mu) + \mu(x/\theta_R), & \theta_F < x \leq \theta_R \\ 1, & x > \theta_R \end{cases} \quad \text{where } \frac{1}{\tilde{\theta}} \equiv \frac{1 - \mu}{\theta_F} + \frac{\mu}{\theta_R} \quad (8)$$

Below θ_F , there are two possibilities of the market tolerance, and firms can't tell what state they're in. Thus, they act off an *effective ceiling* of the probability weighted tolerance thresholds. We denote this by $\tilde{\theta}$, which happens to be the probability weighted harmonic mean of θ_F and θ_R . Since $1/\tilde{\theta}$ is a weighted average of $1/\theta_R$ and $1/\theta_F$, it lies in between them, giving us

$$\theta_F < \tilde{\theta} < \theta_R \quad (9)$$

Above θ_F , every fragile market consumer has been lost, so the effective ceiling drops from $1/\tilde{\theta}$ to just μ/θ_R as only robust consumers are left to lose. A useful property of G is that if we multiply θ_F and θ_R by $t > 0$, the new CDF G^t satisfies $G^t(tx) = G(x)$, giving us the same scale invariance result as Proposition 3. This allows us to normalize $\theta_F = 1$ so the equilibrium depends only on κ , μ and γ .

5.3. The cautious equilibrium

If both firms stay at or below θ_F , then $G(x) = x/\tilde{\theta}$. In this case, the game collapses to the complete information game where the known θ equals $\tilde{\theta}$. Let F_θ denote the equilibrium of Section 4.4 for a ceiling θ , with support $[\underline{s}, \bar{s}]$ and profit $\pi^*(\theta)$.

Proposition 4 (Cautious equilibrium). *Suppose that:*

1. $\bar{s} \leq \theta_F$, and
2. $\mu \left(r(d) \frac{\theta_R - d}{\theta_R} \right) \leq \pi^*(\tilde{\theta})$ for all $d \in (\theta_F, \theta_R]$

Then both firms drawing from $F_{\tilde{\theta}}$ is a mixed strategy Nash equilibrium. Each firm earns $\pi^*(\tilde{\theta}) = \tilde{\theta}^\gamma \pi^*(1)$, and every deception level is the complete information level with ceiling $\theta = 1$ scaled up by $\tilde{\theta}$.

Proof. There are three possible regions in which a firm can deviate. We can compute payoffs in each region using G by using Lemma 1.

Case 1: $d \leq \theta_F$. We have $G = x/\tilde{\theta}$ and the result that each firm would earn the same as in the complete information game with the ceiling $\tilde{\theta}$. $F_{\tilde{\theta}}$ is an equilibrium of this game, and never exceeds $\pi^*(\tilde{\theta})$, with equality on $[\underline{s}, \bar{s}]$.

Case 2: $\theta_F < d \leq \theta_R$. Since it deviates to be the high firm, they sell $1 - G(d) = \mu \left(\frac{\theta_R - d}{\theta_R} \right)$, which never exceeds $\pi^*(\tilde{\theta})$ by the second assumption.

Case 3: $d > \theta_R$. The firm exceeds everyone's tolerance thresholds, earning 0.

Thus, no deviation is profitable, so $F_{\tilde{\theta}}$ is a mixed strategy Nash equilibrium if the assumptions hold. The scaling results follow from Proposition 3. $\square$

The first condition states that the upper end of the support never exceeds θ_F , so the cautious equilibrium never deceives past what the fragile market would tolerate. The second tells us that it is never profitable to jump past θ_F , despite the firm being the sole player in the region. For $r(d) = d^\gamma$, the monopoly profit is maximized at $d^{\text{mon}}(\theta_R) = \frac{\gamma}{\gamma+1} \theta_R$ (Section 3.1). Thus, the second condition can be rewritten as $\mu \pi^{\text{mon}}(\theta_R) \leq \pi^*(\tilde{\theta})$, which tells us that gambling on the market being robust must pay less than staying cautious.

5.4. Uncertainty restrains deception

We compare the cautious equilibrium with two benchmarks: an average market in which the firms know the ceiling is $\mathbb{E}[\theta] = (1 - \mu)\theta_F + \mu\theta_R$, and full information, in which firms know θ .

Proposition 5. *Under the assumptions of Proposition 4, we observe that:*

- (i) *Every deception level is strictly lower than in the average market, and expected deception is strictly lower than under full information.*
- (ii) *Firms deceive more than if they knew the market was fragile, and less than if they knew it was robust.*
- (iii) *Expected profit is strictly lower than under full information.*

Proof. (i) Consider $\frac{\mathbb{E}[\theta]}{\tilde{\theta}} = [(1 - \mu)\theta_F + \mu\theta_R] \left[\frac{1 - \mu}{\theta_F} + \frac{\mu}{\theta_R} \right] = (1 - \mu)^2 + \mu^2 + \mu(1 - \mu) \left(\kappa + \frac{1}{\kappa} \right)$.
As $1 = ((1 - \mu) + \mu)^2 = (1 - \mu)^2 + \mu^2 + \mu(1 - \mu)$,

$$\frac{\mathbb{E}[\theta]}{\tilde{\theta}} - 1 = \mu(1 - \mu) \left(\kappa + \frac{1}{\kappa} - 2 \right) = \mu(1 - \mu) \frac{(\kappa - 1)^2}{\kappa} > 0$$

Thus, $\tilde{\theta} < \mathbb{E}[\theta]$. We also know that levels at ceiling 1 are strictly positive as we need $\underline{s}(1) > 0$ to ensure a strictly positive revenue and profit. Hence, by Proposition 3 and 4, the mean and

support endpoints are strictly lower, since each equals $\tilde{\theta}$ or $\mathbb{E}[\theta]$ times the corresponding ceiling at 1. A more general argument shows that since $d/\tilde{\theta} > d/\mathbb{E}[\theta]$,

$$F_{\tilde{\theta}}(d) = F_1\left(\frac{d}{\tilde{\theta}}\right) \geq F_1\left(\frac{d}{\mathbb{E}[\theta]}\right) = F_{\mathbb{E}[\theta]}(d)$$

Thus, for every d , firms are at least as likely to deceive by at most d , so the whole distribution shifts down. Under full information, the expected deception is $(1 - \mu)\theta_F\mathbb{E}_1[d] + \mu\theta_R\mathbb{E}_1[d] = \mathbb{E}[\theta]\mathbb{E}_1[d]$, which is greater than $\tilde{\theta}\mathbb{E}_1[d]$.

(ii) The same argument with $\theta_F < \tilde{\theta} < \theta_R$ from (9) gives us $F_{\theta_R} \leq F_{\tilde{\theta}} \leq F_{\theta_F}$ for all d .

(iii) Using the weighted AM-GM, $(1 - \mu)a + \mu b \geq a^{1-\mu}b^\mu$, $a, b > 0$. If $a = 1/\theta_F$ and $b = 1/\theta_R$, then $\tilde{\theta}^{-1} > \theta_F^{-(1-\mu)}\theta_R^{-\mu}$, so $\tilde{\theta} < \theta_F^{1-\mu}\theta_R^\mu$. Raising both sides to the power γ and using weighted AM-GM again, we get

$$\tilde{\theta}^\gamma < (\theta_F^\gamma)^{1-\mu}(\theta_R^\gamma)^\mu < (1 - \mu)\theta_F^\gamma + \mu\theta_R^\gamma$$

Multiplying the ends of the inequality by $\pi^*(1)$, we get $\pi^*(\tilde{\theta}) < (1 - \mu)\pi^*(\theta_F) + \mu\pi^*(\theta_R)$. Thus, expected profit under uncertainty (which is always $\pi^*(\tilde{\theta})$ by Proposition 4) is lower than expected profit under full information. $\square$

5.5. Overreach and busts

Now let's look at when firms start to gamble in equilibria, i.e. they play $d > \theta_F$ with a positive probability. We say a firm *overreaches* if $d > \theta_F$, and a firm that overreaches *busts* if the market turns out to be fragile and thus, a firm that busts earns nothing. Say a firm overreaches with probability $p \equiv P(d > \theta_F)$. Since firms draw independently, the probability that neither overreaches is $(1 - p)^2$, so at least one overreaches with probability $1 - (1 - p)^2$. Thus, the probability at least one firm busts is

$$P(\text{bust}) = (1 - \mu)[1 - (1 - p)^2] \quad (10)$$

Repeating the steps from Section 4 with G instead of the uniform CDF, we find the density of the support is

$$f(d) = \frac{1}{1 - G(d)} \left[g(d) - \frac{r'(d)\pi^*}{r(d)^2} \right]$$

with the endpoints satisfying

$$\pi^* = \frac{g(\underline{s})r(\underline{s})^2}{r'(\underline{s})}, \quad \pi^* = r(\bar{s})[1 - G(\bar{s})]$$

We can solve this numerically to check that no $d \in (0, \theta_R)$ earns more than π^* on a grid. Besides the already established cautious equilibrium, we find three new equilibrium types:

(i) *Overreach*: The support is a single interval $[\underline{s}, \bar{s}]$ that crosses θ_F .

(ii) *Split*: The support is split in two, with $[s, s_1] \cup [s_2, \bar{s}]$, separated around θ_F .

(iii) *All in*: The whole support lies above θ_F .

Why does a split equilibrium have a gap around θ_F while the overreach support runs right through it? Raising deception always comes with a cost and a benefit. The cost is that you lose consumers, which happens at an expected loss rate $g(d)$. The benefit is the gain in revenue, which is $\frac{r'(d)}{r(d)} \cdot D(d)$, where $D(d)$ is expected demand from (4). Since $D(d) = \pi^*/r(d)$, the gain in revenue is $\frac{r'(d)}{r(d)^2} \pi^* = \gamma \pi^*/d^{\gamma+1}$. Therefore, firms play d only if $g(d) \geq \gamma \pi^*/d^{\gamma+1}$. The gain falls smoothly as d rises, but the cost is piecewise as

$$g(d) = \begin{cases} 1/\tilde{\theta}, & d < \theta_F \\ \mu/\theta_R, & d \geq \theta_F \end{cases}$$

The cost is kinked at θ_F , causing a sudden discontinuity in the firm's choice. Whether a firm plays θ_F in the support is determined by the sign of $\frac{\gamma \pi^*}{\theta_F^{\gamma+1}} - \frac{\mu}{\theta_R}$, i.e. the revenue gain at θ_F minus the loss rate there. If it is negative, then firms play θ_F , giving us the overreach equilibrium. If it is positive, then there's more profit to be made above θ_F , so it is not part of the support, giving us a split equilibrium.

Table 2: Equilibrium under uncertainty ($\gamma = 1$, $\theta_F = 1$)

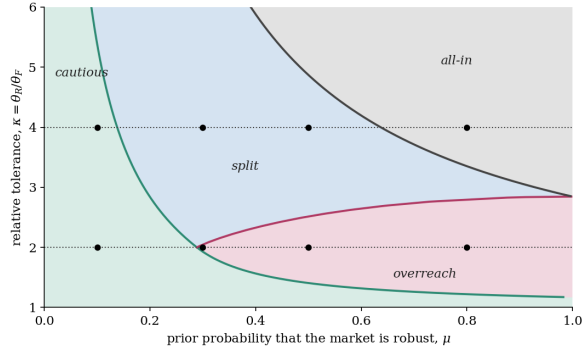
κ	μ	Shape	Support	$\mathbb{E}[d]$	$\Pr(d > \theta_F)$	$\Pr(\text{bust})$
2	0.1	cautious	[0.370, 0.901]	0.740	0	0
2	0.3	overreach	[0.414, 1.171]	0.831	0.033	0.045
2	0.5	overreach	[0.481, 1.553]	1.069	0.466	0.357
2	0.8	overreach	[0.600, 1.678]	1.314	0.801	0.192
4	0.1	cautious	[0.380, 0.925]	0.760	0	0
4	0.3	split	[0.504, 0.949] $\cup$ [1.621, 3.171]	1.877	0.573	0.572
4	0.5	split	[0.651, 0.923] $\cup$ [1.455, 3.372]	2.580	0.896	0.495
4	0.8	all-in	[1.406, 3.423]	2.811	1	0.200

Busts. The risk of a bust is hump-shaped in μ (Figure 2b). It is zero up to the cautious threshold, below which firms never overreach, and it vanishes as $\mu \rightarrow 1$, because a fragile market then almost never occurs. In between, firms gamble and busts are common. Risk peaks at $\mu \approx 0.52$, where it reaches 0.36, when $\kappa = 2$, and at $\mu \approx 0.34$, where it reaches 0.58, when $\kappa = 4$.

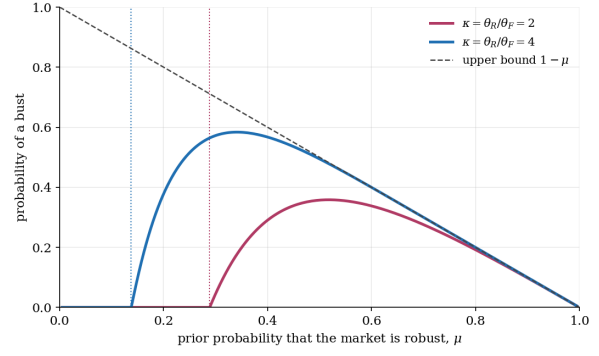
5.6. Summary

When firms do not know how much deception customers will tolerate, what they do depends on how likely a tolerant market is. If it is unlikely enough, uncertainty makes firms careful: they act as if customers were less tolerant than they are on average, because customers are lost fastest in a fragile market. Deception is then lower on average than if firms knew the market, and so are profits.

Once a tolerant market is likely enough, firms gamble. Each sometimes pushes deception past what a fragile market accepts, and when the market turns out to be fragile, every consumer



(a) Equilibrium shape; dots mark the rows of Table 2



(b) Probability of a bust; dotted lines mark the end of the cautious region

Figure 2: Equilibrium under uncertainty across beliefs μ and relative tolerance $\kappa = \theta_R/\theta_F$ ($\gamma = 1$, $\theta_F = 1$).

rejects the overreaching firm and it sells nothing. These busts are not mistakes but the downside of a bet worth making on average. They are most likely at intermediate odds, when a tolerant market is neither unlikely nor close to certain.

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